HDOC – Experiment 5

Question 1

Write a C program to find the area, arc length and perimeter in meters of a quadrant when its radius is given in centimetres.

❖ Given requirements

INPUT

Radius of quadrant in centimeters (cm)

OUTPUT

- Area of quadrant (m^2)

- Arc Length (m)

- Perimeter (m)

❖ Formula

Radius into meters:	Radius(m) = Radius(cm)/100
Area of quadrant:	Area= $\pi r^2/4$
Arc length:	Arc= $\pi r/2$
Perimeter:	Perimeter= $2r+ \pi r/2$

❖ Test case table

Test case	Input(cm)	Expected output	Actual output	Result
1	100	Area= 0.785 m^2 Arc= 1.570m Perimeter=3.570m	Same	Pass
2	50	Area=0.196m^2 Arc=0.785m Perimeter=1.785m	Same	Pass
3	200	Area=3.141m^2 Arc=3.141m Perimeter=7.141m	Same	Pass

❖ Algorithm

1. Start
2. Declare variables for radius, area, arc and perimeter.
3. Read radius in centimetres.
4. Convert radius into meters
5. Calculate area of quadrant.
6. Calculate arc length.
7. Calculate perimeter.
8. Display all results.
9. Stop

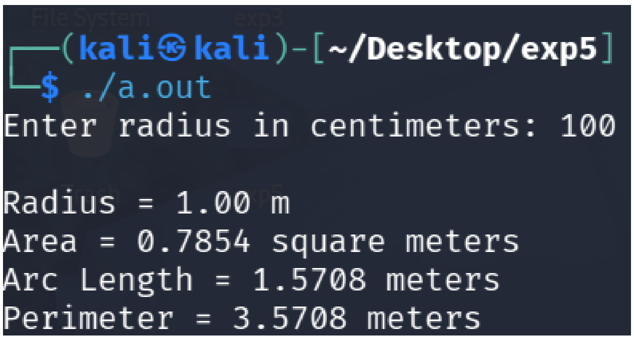
❖ Program code in C

```
#include <stdio.h>

#define PI 3.141592653589793

int main()

{

  double radius_cm, radius_m;

  double area, arcLength, perimeter;
```

```c
    printf("Enter radius in centimeters: ");
    scanf("%lf", &radius_cm);

    radius_m = radius_cm / 100.0;

    area = (PI * radius_m * radius_m) / 4.0;
    arcLength = (PI * radius_m) / 2.0;
    perimeter = (2 * radius_m) + arcLength;

    printf("\nRadius = %.2f m\n", radius_m);
    printf("Area = %.4f square meters\n", area);
    printf("Arc Length = %.4f meters\n", arcLength);
    printf("Perimeter = %.4f meters\n", perimeter);

    return 0;
}
```

❖ Compile, Execute and Evaluate

TEST CASE 1:



TEST CASE 2:

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter radius in centimeters: 50

Radius = 0.50 m
Area = 0.1963 square meters
Arc Length = 0.7854 meters
Perimeter = 1.7854 meters
```

TEST CASE 3:

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter radius in centimeters: 200

Radius = 2.00 m
Area = 3.1416 square meters
Arc Length = 3.1416 meters
Perimeter = 7.1416 meters
```

EVALUATION: All the three cases produces the expected results, therefore, the program passes all the cases.

❖ Compilation and Execution screenshot

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ vi quadrant.c

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ cc quadrant.c

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter radius in centimeters: 100

Radius = 1.00 m
Area = 0.7854 square meters
Arc Length = 1.5708 meters
Perimeter = 3.5708 meters

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter radius in centimeters: 50

Radius = 0.50 m
Area = 0.1963 square meters
Arc Length = 0.7854 meters
Perimeter = 1.7854 meters

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ 200
200: command not found

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter radius in centimeters: 200

Radius = 2.00 m
Area = 3.1416 square meters
Arc Length = 3.1416 meters
Perimeter = 7.1416 meters
```

Question 2

Write a C program to find the total area and total boundary length of a rectangle with two semicircular ends when the rectangle length, breadth, and semicircle radius are given.

❖ Given requirements

INPUT

- Rectangle Length (m)
- Rectangle Breadth (m)
- Radius of semicircle (m)

OUTPUT

- Total Area
- Total Boundary Length

❖ Formula

Area of rectangle: $Area_{rectangle} = Length \times Breadth$

Area of two Semicircles: $Area_{circle} = \pi r^2$

Total Area: $Area = Length \times Breadth + \pi r^2$

Straight sides: $2 \times Length$

Curved path: $2\pi r$

Total boundary length: $Boundary = 2Length + 2\pi r$

❖ Test case table

Test case	Input(L,B,R)	Expected output	Actual output	Result
1	10, 4, 2	Area=52.566m^2 Boundary=32.5664m	SAME	PASS
2	8, 3, 1.5	Area=31.068m^2 Boundary=25.424m	SAME	PASS
3	15, 6, 3	Area=118.274m^2 Boundary=48.849m	SAME	PASS

❖ Algorithm

1. Start
2. Declare length, breadth and radius.
3. Calculate rectangle area.
4. Calculate area of circle.
5. Add both areas.
6. Calculate boundary length.
7. Display total area and boundary length.
8. Stop

❖ Program code in C

```
#include <stdio.h>

#define PI 3.141592653589793

int main()

{
```

```
    double length, breadth, radius;
    double totalArea, boundaryLength;

    printf("Enter rectangle length: ");
    scanf("%lf", &length);

    printf("Enter rectangle breadth: ");
    scanf("%lf", &breadth);

    printf("Enter semicircle radius: ");
    scanf("%lf", &radius);

    totalArea = (length * breadth) + (PI * radius * radius);

    boundaryLength = (2 * length) + (2 * PI * radius);

    printf("\nTotal Area = %.4f square meters\n", totalArea);
    printf("Total Boundary Length = %.4f meters\n", boundaryLength);

    return 0;
}
```

❖ Compile, Execute and Evaluate

Test case 1:

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 10
Enter rectangle breadth: 4
Enter semicircle radius: 2

Total Area = 52.5664 square meters
Total Boundary Length = 32.5664 meters
```

Test case 2:

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 8
Enter rectangle breadth: 3
Enter semicircle radius: 1.5

Total Area = 31.0686 square meters
Total Boundary Length = 25.4248 meters
```

Test case 3:

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 15
Enter rectangle breadth: 6
Enter semicircle radius: 3

Total Area = 118.2743 square meters
Total Boundary Length = 48.8496 meters
```

EVALUATION: All the three cases produce the expected results. Therefore, the program passes all the cases.

❖ Compilation and Execution screenshot

```
┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ vi rectangle.c

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ cc rectangle.c

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 10
Enter rectangle breadth: 4
Enter semicircle radius: 2

Total Area = 52.5664 square meters
Total Boundary Length = 32.5664 meters

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 8
Enter rectangle breadth: 3
Enter semicircle radius: 1.5

Total Area = 31.0686 square meters
Total Boundary Length = 25.4248 meters

┌──(kali㉿kali)-[~/Desktop/exp5]
└─$ ./a.out
Enter rectangle length: 15
Enter rectangle breadth: 6
Enter semicircle radius: 3

Total Area = 118.2743 square meters
Total Boundary Length = 48.8496 meters
```